IC TRAINING CENTER VIET NAM
FUNDAMENTAL IC DESIGN AND VERIFICATION COURSE



IC DESIGN AND VERIFICATION

Final Project:

Timer IP

Class: IC25

Instructor: Nguyen Duc Tri
Student: Huynh Minh Nhat

HO CHI MINH CITY, September 2025

Contents

1 Overview **2**
1.1 Introduction . . . 2
1.2 Main features . . . 2
1.3 Block diagram . . . 2
1.4 Interface signals . . . 3

2 Register specification **3**
2.1 Register summary . . . 3
2.2 Timer Control Register (TCR) . . . 4
2.3 Timer Data Register 0 – TDR0 . . . 4
2.4 Timer Data Register 1 – TDR1 . . . 4
2.5 Timer Compare Register 0 – TCMP0 . . . 4
2.6 Timer Compare Register 1 – TCMP1 . . . 4
2.7 Timer Interrupt Enable Register – TIER . . . 5
2.8 Timer Interrupt Status Register – TISR . . . 5
2.9 Timer Halt Control Status Register – THCSR . . . 5

3 Functional description **6**
3.1 APB Slave . . . 6
3.2 Reg file . . . 6
3.2.1 TCR Register . . . 7
3.2.2 TDR0/1 Register . . . 7
3.2.3 TCMP0/1 Register . . . 9
3.2.4 TIER Register . . . 9
3.2.5 TISR Register . . . 10
3.2.6 THCSR Register . . . 10
3.2.7 Pslave error signal . . . 11
3.3 Counter Control . . . 11
3.4 Interrupt . . . 12

4 Coding Path name **12**

5 Testing and Verification **13**
5.1 Summary check . . . 13
5.2 Coverage . . . 13
5.3 Exclude Check . . . 15

1 Overview

1.1 Introduction

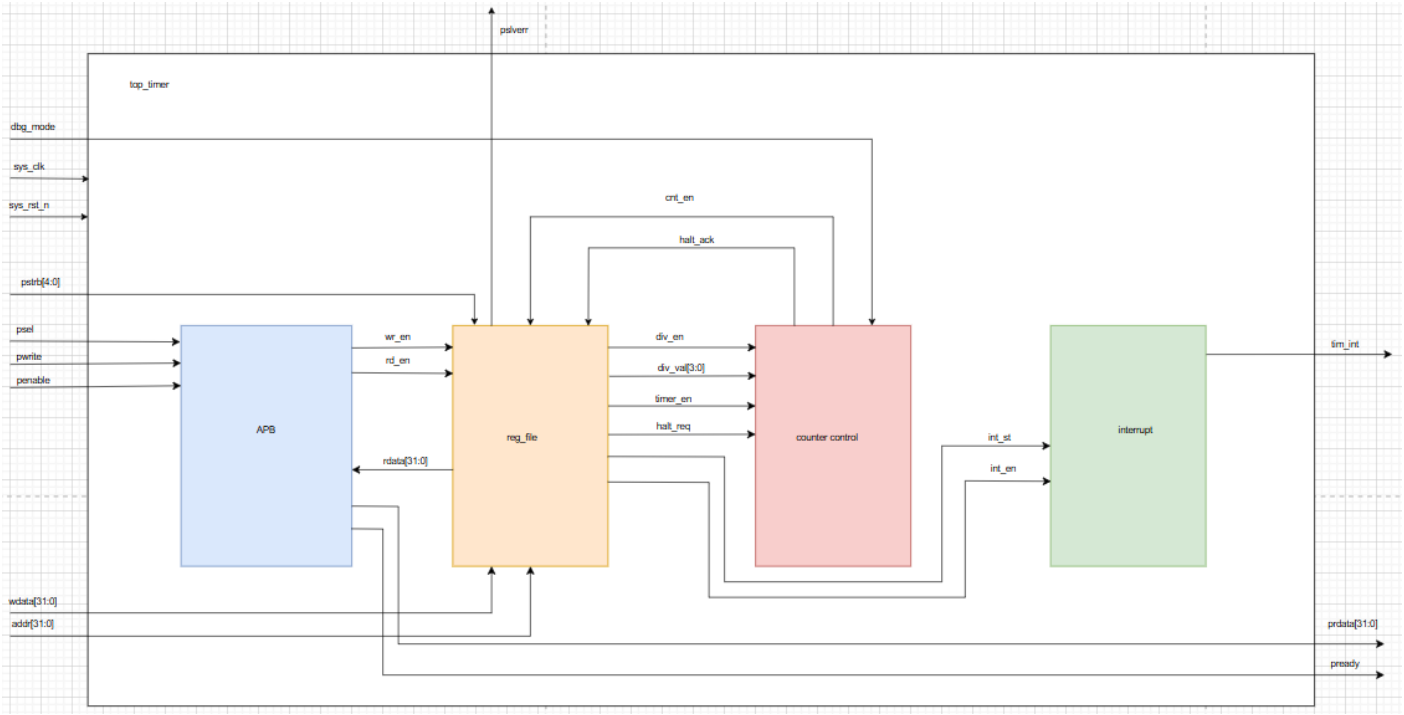
Timer is an essential module for every chip. This is used to generate accurate timing intervals or controlling the timing of various operations within the circuit.

Timer can be used in various application: pulse generation, delay generation, event generation, PWM generation, Interrupt generation .... In this project, a timer module is customized from CLINT module of industrial RISC-V architecture. It is used to generate interruption based on user settings.

1.2 Main features

- 64-bit count-up
- 12-bit address
- Register set is configured via APB bus (IP is APB slave)
- Standard level: Only support APB 32-bit transfer with no wait states and no error handling
- **Advanced level:**
- Support wait state (1 cycle is enough) and error handling
- Support byte access
- Support halt (stop) in debug mode
- System clock frequency is 200 MHz. Timer uses active low async reset
- Counter can be counted based on system clock or divided up to 256
- Support timer interrupt (can be enabled or disabled)

1.3 Block diagram



Hình 1: Block Diagram

1.4 Interface signals

Signal name	Width	Direction	Description
sys_clk	1	input	System clock frequency is 200 MHz -> 5 ns/cycle
sys_rst_n	1	input	Active low asynchronous reset. 0: counter is in reset state 1: counter is in non-reset s
tim_psel	1	input	when it is high, pwdata is written corresponding with
tim_pwrite	1	input	write when = 1, read when = 0
tim_penable	1	input	Goes high one cycle after PSEL, remains high until transfer compl
tim_paddr[11:0]	12	input	register's address, from 0x4000_1000 -> 0x4000_1FFF = 4096 Bytes = 1024 registers for 32(bit) = 512 registers for 64(bit
tim_pwdata[31:0]	32	input	data will be stored in register
tim_prdata[31:0]	32	output	data will be loaded immediately (pready = 1) from register
tim_pstrb[3:0]	4	input	indicate index that data will be stored in what is bit in a register
tim_pready	1	output	ready for reading data
tim_pslverr	1	output (slave)	Asserted HIGH in the last cycle of a transfer if an error response is generated
tim_int	1	outpu	Timer interrupt (tim_int) is asserted (set) when interrupt is enabled and counter's value m (equal) the compare value.
dbg_mode	1	input	For simple, dbg_mode does not change after timer_en is High.

Hình 2: IO Portlist

2 Register specification

2.1 Register summary

Base address: 0x4000_1000

Offset	Abbreviation	Register name
0x00	TCR	Timer Control Register
0x04	TDR0	Timer Data Register 0
0x08	TDR1	Timer Data Register 1
0x0C	TCMP0	Timer Compare Register 0
0x10	TCMP1	Timer Compare Register 1
0x14	TIER	Timer Interrupt Enable Register
0x18	TISR	Timer Interrupt Status Register
0x1C	THCSR	Timer Halt Control Status Register
Others	Reserved	

Hình 3: Base address

2.2 Timer Control Register (TCR)

Bit	Name	Type	Default value	Description
31:12	Reserved	-	20'h0	Reserved
11:8	div_val	RW	4'b0001	Counter control mode setting: • 4'b0000: Counting speed is not divided • 4'b0001: Counting speed is divided by 2 (default) • 4'b0010: Counting speed is divided by 4 • 4'b0011: Counting speed is divided by 8 • 4'b0100: Counting speed is divided by 16 • 4'b0101: Counting speed is divided by 32 • 4'b0110: Counting speed is divided by 64 • 4'b0111: Counting speed is divided by 128 • 4'b1000: Counting speed is divided by 256 • Others: reserved, (*)prohibit settings. When setting the prohibit value, div_val is not changed. Note: user must not change div_en while timer_en is High (*): add hardware logic to ensure div_val is prohibited to change when timer_en is High. Access is error response in this case. (*)access is "error response" when setting prohibit value to div_val (*): advanced level
7:2	Reserved	RO	6'b0	Reserved
1	div_en	RW	1'b0	Counter control mode enable. • 0: Disabled. Counter counts with normal speed based on system clock • 1: Enabled. The couting speed of counter is controlled based on div_val Note: user must not change div_en while timer_en is High (*): add hardware logic to ensure div_en is prohibited to change when timer_en is High. Access is error response in this case.
0	timer_en	RW	1'b0	Timer enable • 0: Disabled. Counter does not count. • 1: Enabled. Counter starts counting. (*) timer_en changes from H->L will initialize the TDR0/1 to their initial value

Hình 4: Width of Elements in TCR Register

2.3 Timer Data Register 0 – TDR0

Bit	Name	Type	Default value	Description
31:0	TDR0	RW	32'h0000_0000	Lower 32-bit of 64-bit counter. [Advanced level]: value of this register is cleared to initial value when timer_en changes from H->L.

Hình 5: Width of Elements in TDR0 Register

2.4 Timer Data Register 1 – TDR1

Bit	Name	Type	Default value	Description
31:0	TDR1	RW	32'h0000_0000	Upper 32-bit of 64-bit counter. [Advanced level]: value of this register is cleared to initial value when timer_en changes from H->L.

Hình 6: Width of Elements in TDR1 Register

2.5 Timer Compare Register 0 – TCMP0

Bit	Name	Type	Default value	Description
31:0	TCMP0	RW	32'hFFFF_FFFF	Lower 32-bit of 64-bit compare value. Interrupt is asserted when counter value is qual to compare value.

Hình 7: Width of Elements in TCMP0 Register

2.6 Timer Compare Register 1 – TCMP1

Bit	Name	Type	Default value	Description
31:0	TCMP1	RW	32'hFFFF_FFFF	Upper 32-bit of 64-bit compare value. Interrupt is asserted when counter value is qual to compare value.

Hình 8: Width of Elements in TCMP1 Register

2.7 Timer Interrupt Enable Register – TIER

Bit	Name	Type	Default value	Description
31:1	Reserved	RO	31'h0	Reserved
0	int_en	R/W	1'b0	Timer interrupt enable 0: Timer interrupt is disabled. 1: Timer interrupt is enabled. When this bit is 0, no timer interrupt is output. When this bit is 1, timer interrupt can be output when reaching trigger condition. Clearing this bit to 0 while interrupt is asserting will mask the interrupt to 0 but does not affect the interrupt pending bit TISR.int_st bit

Hình 9: Width of Elements in TIER Register

2.8 Timer Interrupt Status Register – TISR

Bit	Name	Type	Default value	Description
31:1	Reserved	RO	31'h0	Reserved
0	int_st	RW1C	1'b0	Timer interrupt trigger condition status bit (interrupt pending bit) 0: the interrupt trigger condition does not occur. 1: the interrupt trigger condition occurred. Write 1 when this bit is 1 to clear it Write 0 when this bit is 1 has no effect Write to this bit when it is 0 has no effect. Note: When interrupt trigger condition occurred (counter reached compare value), counter continues to count normally.

Hình 10: Width of Elements in TISR Register

2.9 Timer Halt Control Status Register – THCSR

Bit	Name	Type	Default value	Description
31:2	Reserved	RO	30'h0	Reserved
1	halt_ack	RO	1'b0	[Standard level] This bit is reserved bit [Advanced level] Timer halt acknowledge 0: timer is NOT halted 1: timer is halted Timer accepts the halt request only in debug mode, indicates by debug_mode input signal
0	halt_req	RW	1'b0	[Standard level] This bit is normal R/W but has no function related. [Advanced level] Timer halt request 0: no halt req. 1: timer is requested to halt.

Hình 11: Width of Elements in THCSR Register

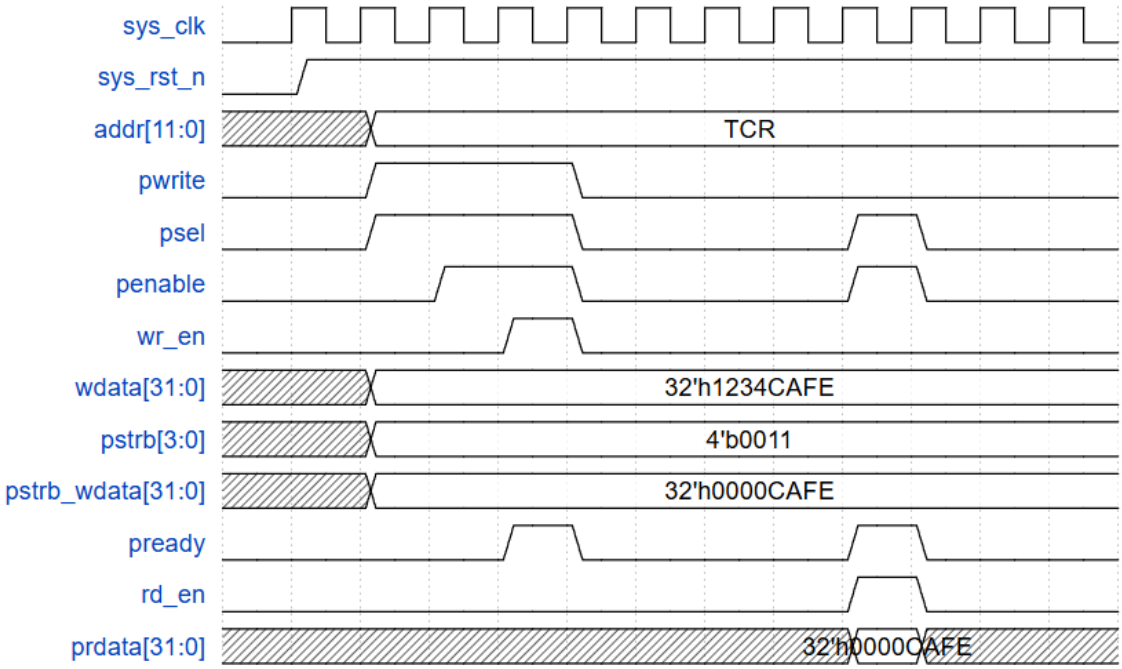
3 Functional description

3.1 APB Slave

The timer IP has an APB slave to receive settings from user. This APB slave has below features:

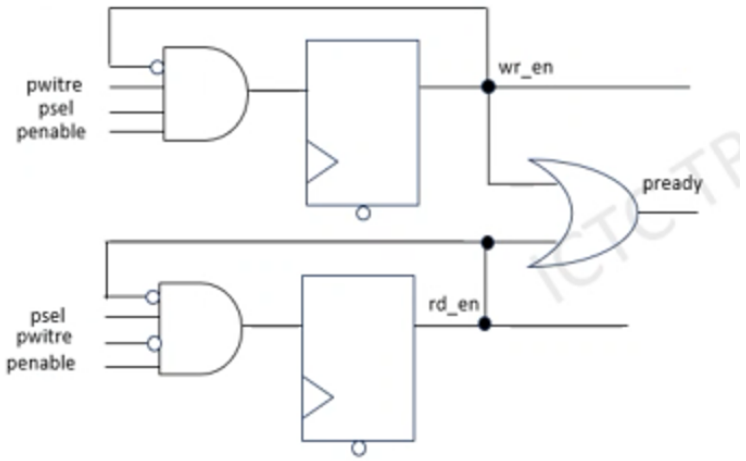
- 12-bit width address
- 32-bit width read/write data
- Supporting wait state and error response

Waveform:



Hình 12: Timing diagram of APB Slave

Logic Diagram:



Hình 13: Logic diagram of APB Slave module

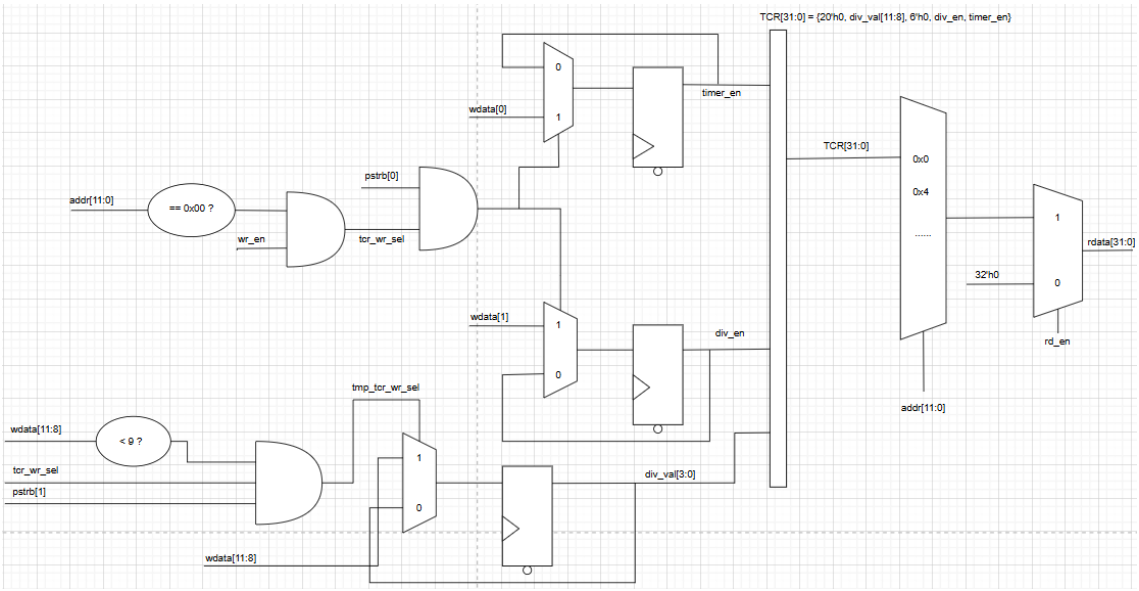
3.2 Reg file

- Stores all control and status registers of the timer IP.
- Handles APB write operations: updates registers on wr_en with address decode.

- Handles APB read operations: returns register contents on rdata based on addr.
- Provides configuration signals (timer_en, div_en, div_val, halt_req) to the Counter Control block.
- Interfaces with the Counter: writes initial/reload values, reads current counter value.
- Manages interrupt registers (TIER, TISR) for enable and status control.
- Ensures only valid updates (e.g., checks field constraints like div_val $<$ 9).
- Supports byte-wise writes using pstrb when APB4 is enabled.

3.2.1 TCR Register

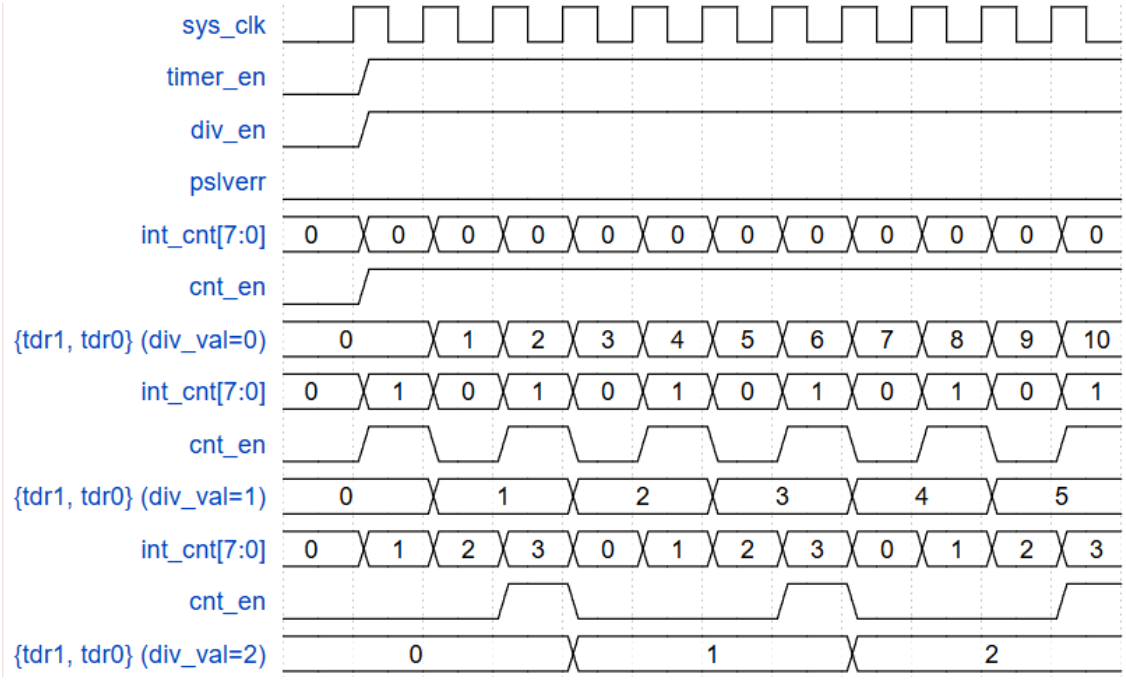
Logic Diagram:



Hình 14: Logic diagram of TCR Register

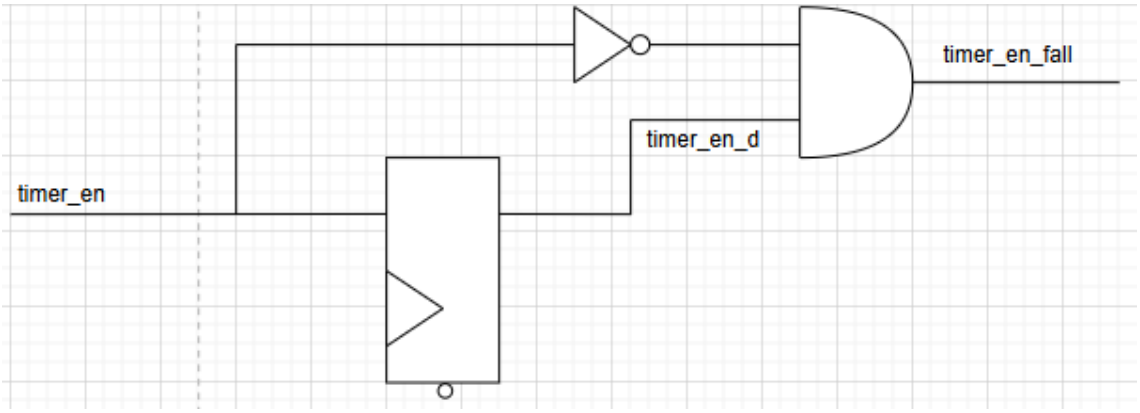
3.2.2 TDR0/1 Register

Waveform:

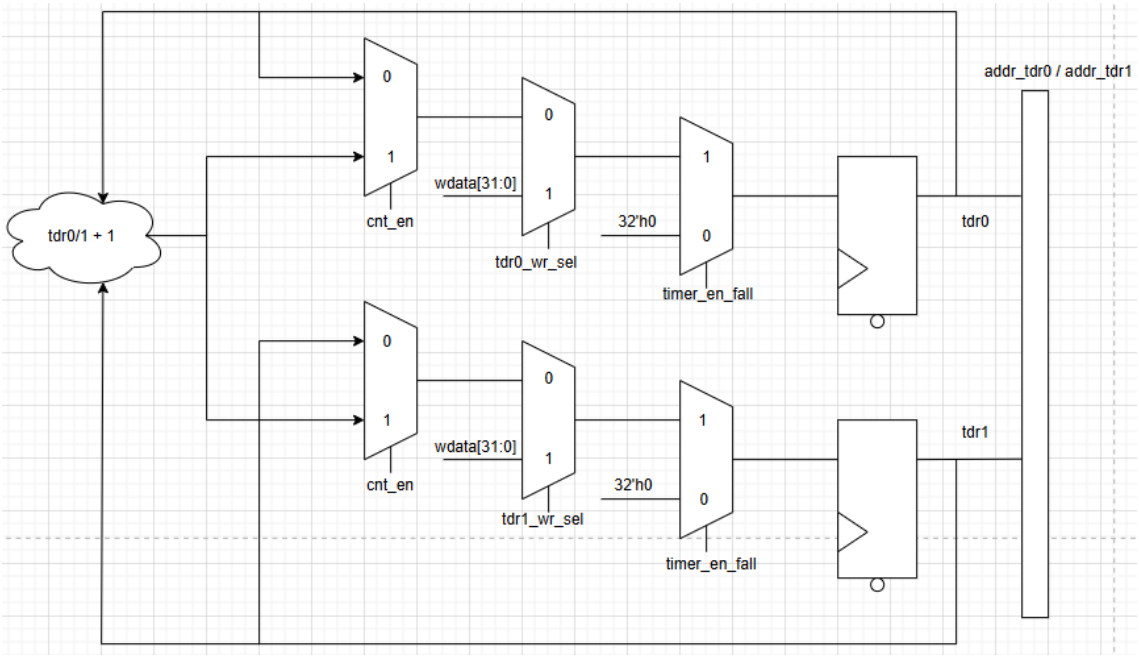


Hình 15: Timing diagram of counting in TDR Register (Counter)

Logic Diagram:



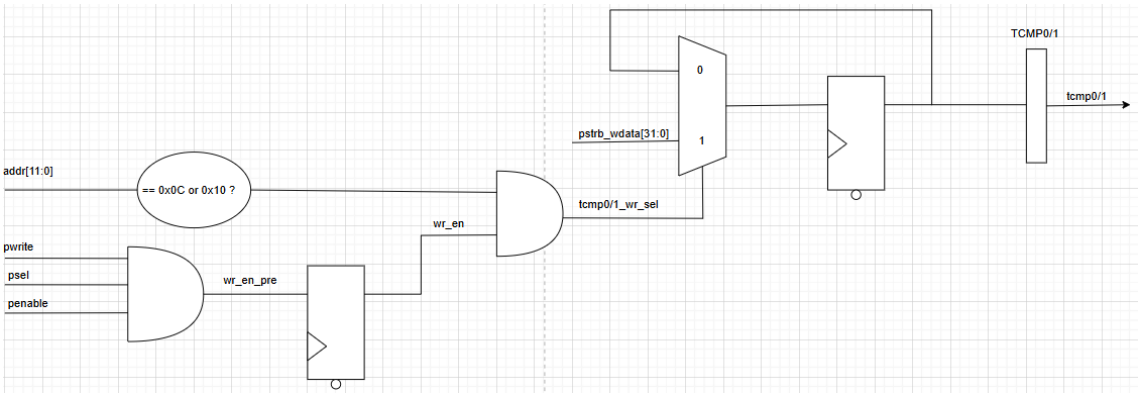
Hình 16: Logic diagram of Detect falling edge of timer_enRegister



Hình 17: Logic diagram of TDR0/1 Register

3.2.3 TCMP0/1 Register

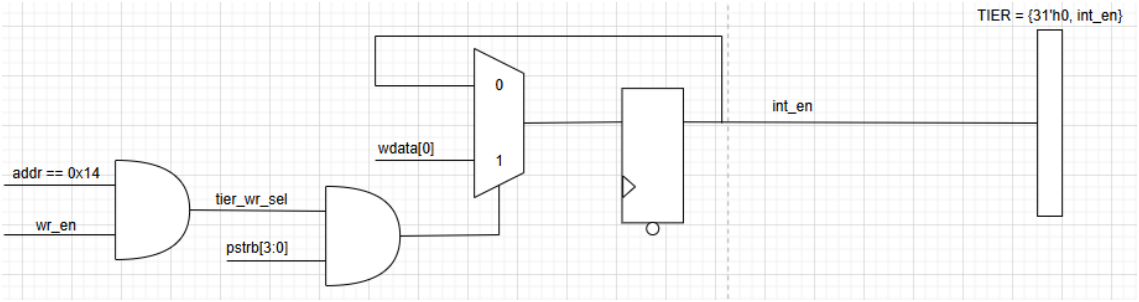
Logic Diagram:



Hình 18: Logic diagram of TCMP0/1 Register

3.2.4 TIER Register

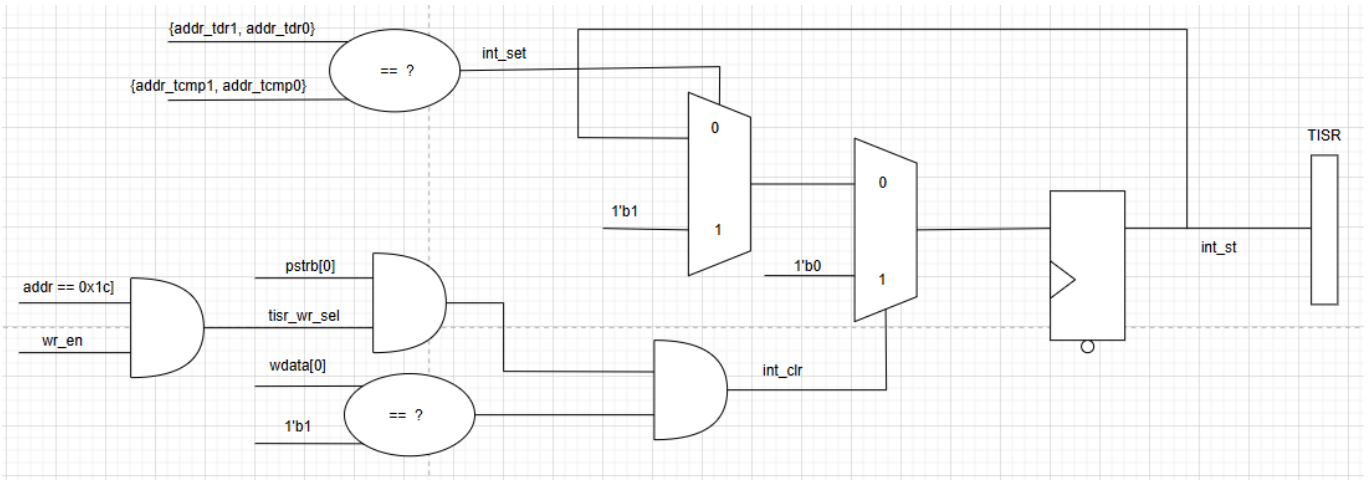
Logic Diagram:



Hình 19: Logic diagram of TIER Register

3.2.5 TISR Register

Logic Diagram:



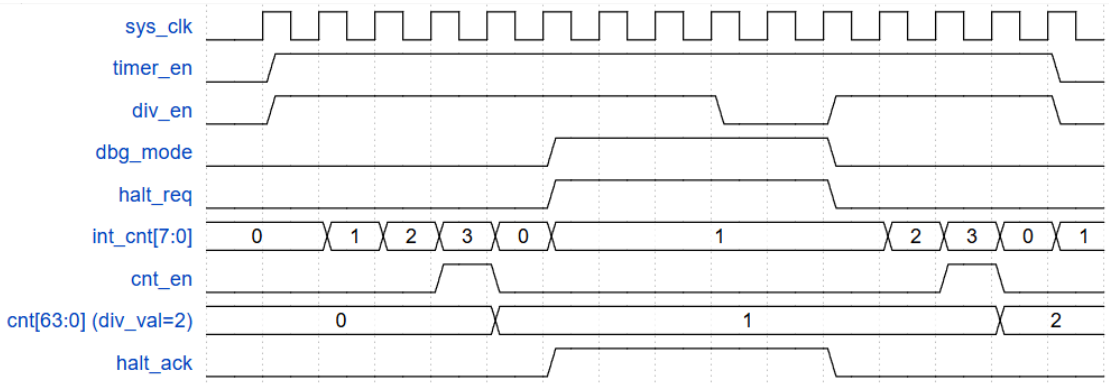
Hình 20: Logic diagram of TISR Register

3.2.6 THCSR Register

Halted

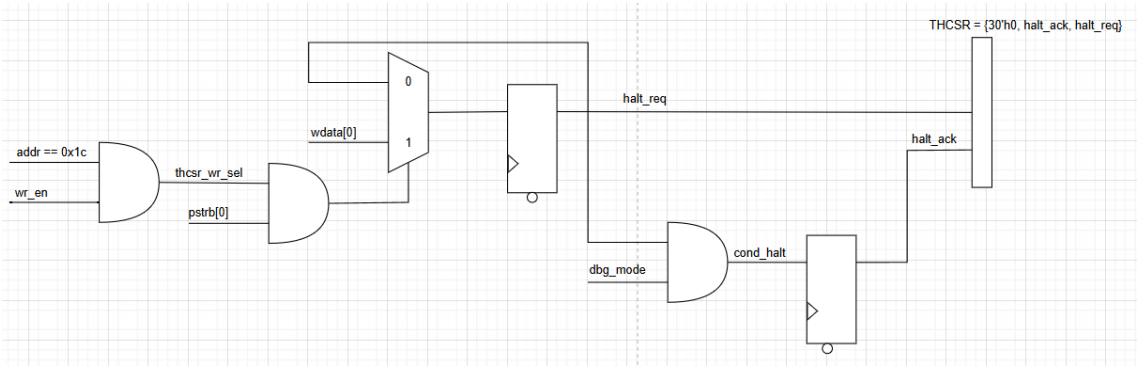
- Counter can be halted (stopped) in debug mode when both below conditions occur:
- Input debug_mode signal is High,
- THCR.halt_req is 1.
- THCSR.halt_ack is 1 after a halt request indicates that the request is accepted.
- After halted, counter can be resumed to count normally when clearing the halt request to 0.
- The period of each counting number needs to be same when halt and not halt as described in the below waveform example (div_val=2).

Waveform of Halt state:



Hình 21: Timing diagram of Halt signal

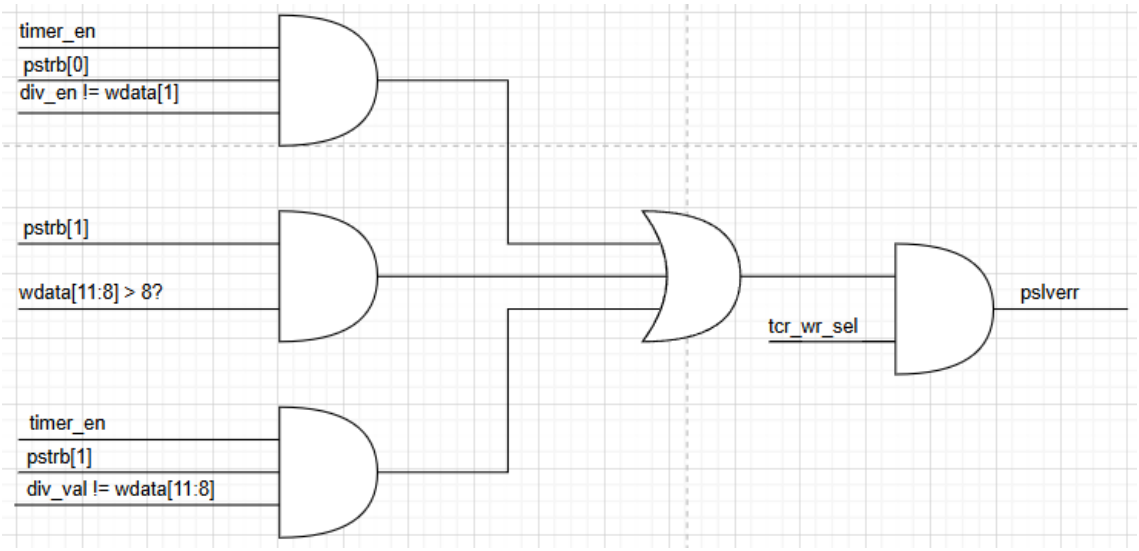
Logic diagram of Halt:



Hình 22: Logic diagram of Halt signal

3.2.7 Pslave error signal

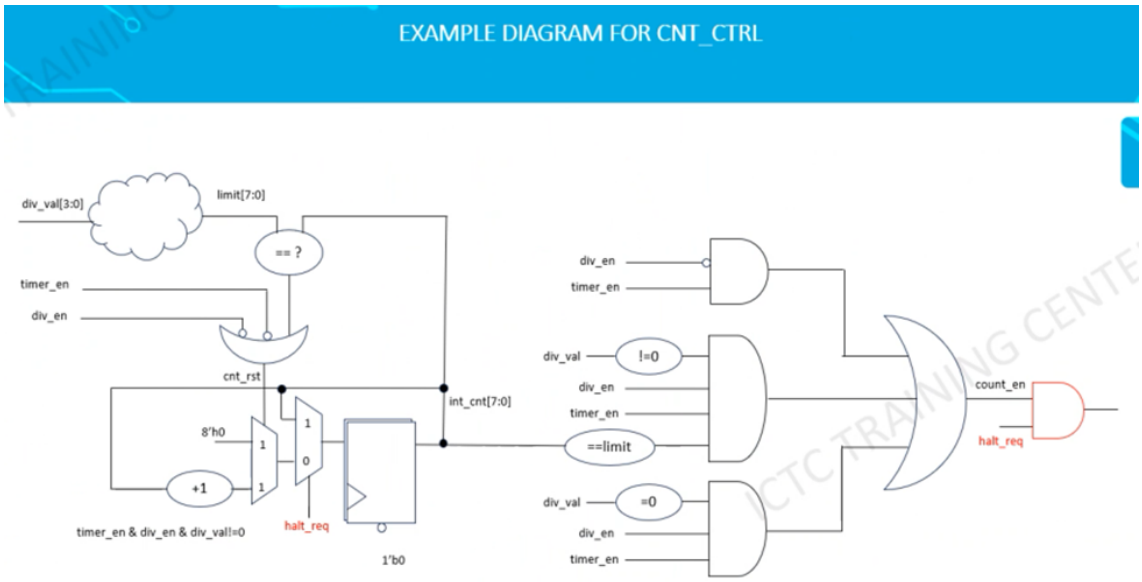
Logic Diagram:



Hình 23: Logic diagram of Pslave error

3.3 Counter Control

Logic Diagram:

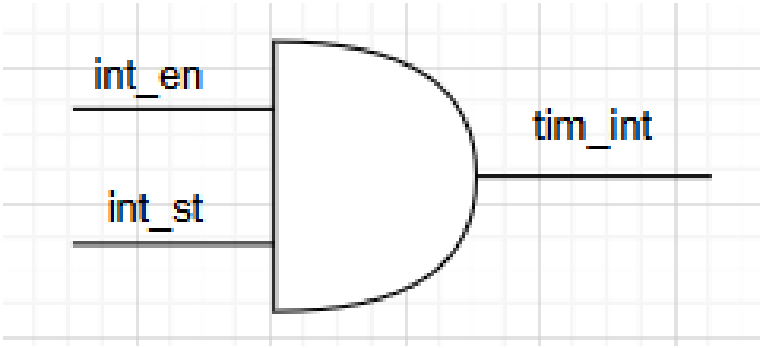


Hình 24: Logic diagram of Counter Control module

3.4 Interrupt

- Timer interrupt (tim_int) is asserted (set) when interrupt is enabled and counter's value matches (equal) the compare value.
- Once asserted, the timer interrupt (tim_int) remains unchange until it is cleared by writing 1 to TISR.int_st bit or the interrupt is disabled.

Logic Diagram:



Hình 25: Logic diagram of Interrupt module

4 Coding Path name

- RTL Code: minhnhat@ictc-eda-ldap:~/final_project/rtl/
- Test bench (Golden model): minhnhat@ictc-eda-ldap:~/final_project/tb/test_bench.v
- Test cases (Golden model): minhnhat@ictc-eda-ldap:~/final_project/testcases/
- Test bench (Self-test): minhnhat@ictc-eda-ldap:~/final_project/my_tb/test_bench.v
- Test cases (Self-test): minhnhat@ictc-eda-ldap:~/final_project/my_testcases/

5 Testing and Verification

5.1 Summary check

PAT_NAME	RUN_DATE	RESULT
reg_init_chk	14:31:52 Sep 28 2025	PASSED
reg_rw_chk	14:31:54 Sep 28 2025	PASSED
reg_reserved_chk	14:31:57 Sep 28 2025	PASSED
reg_1hot_chk	14:31:59 Sep 28 2025	PASSED
reg_byte_access	14:32:01 Sep 28 2025	PASSED
cnt_ctrl_chk	14:32:19 Sep 28 2025	PASSED
apb_protocol_chk	14:32:22 Sep 28 2025	PASSED
apb_multiple_access	14:32:24 Sep 28 2025	PASSED
apb_unaligned_chk	14:32:26 Sep 28 2025	PASSED
cnt_counting_chk	14:32:28 Sep 28 2025	PASSED
interrupt_chk	14:32:30 Sep 28 2025	PASSED
cnt_halt_chk	14:32:46 Sep 28 2025	PASSED
apb_pslverr_chk	14:32:48 Sep 28 2025	PASSED

TOTAL/PASSED/REMAIN:13/13/0

Hình 26: Report summary Pat name check

5.2 Coverage

```
Coverage Report Summary Data by instance

================================================================================
=== Instance: /test_bench/u_timer/u0
=== Design Unit: work.apb_slave
================================================================================
    Enabled Coverage              Bins      Hits    Misses  Coverage
    ----------------              ----      ----    ------  --------
    Branches                         4         4         0   100.00%
    Expressions                     10        10         0   100.00%
    Statements                       7         7         0   100.00%
    Toggles                        148       148         0   100.00%

================================================================================
```

Hình 27: DUT: Apb slave

```
================================================================================
=== Instance: /test_bench/u_timer/u1
=== Design Unit: work.reg_file
================================================================================
    Enabled Coverage              Bins      Hits    Misses  Coverage
    ----------------              ----      ----    ------  --------
    Branches                        91        91         0   100.00%
    Conditions                      10        10         0   100.00%
    Expressions                     34        34         0   100.00%
    Statements                     103       103         0   100.00%
    Toggles                       1054       818       236    77.60%

================================================================================
```

Hình 28: DUT: Reg File

```
================================================================================
=== Instance: /test_bench/u_timer/u2
=== Design Unit: work.counter_ctrl
================================================================================
    Enabled Coverage              Bins      Hits    Misses  Coverage
    ----------------              ----      ----    ------  --------
    Branches                        10         9         1    90.00%
    Expressions                     14        13         1    92.85%
    Statements                      13        13         0   100.00%
    Toggles                         62        61         1    98.38%

================================================================================
```

Hình 29: DUT: Counter Control

```
================================================================================
=== Instance: /test_bench/u_timer/u4
=== Design Unit: work.interrupt
================================================================================
    Enabled Coverage              Bins      Hits    Misses  Coverage
    ----------------              ----      ----    ------  --------
    Expressions                      2         2         0   100.00%
    Statements                       1         1         0   100.00%
    Toggles                         10        10         0   100.00%

================================================================================
```

Hình 30: DUT: Interrupt

```
47 ================================================================================================
48 === Instance: /test_bench/u_timer
49 === Design Unit: work.timer_top
50 ================================================================================================
51     Enabled Coverage              Bins      Hits    Misses  Coverage
52     ----------------              ----      ----    ------  --------
53     Toggles                        268       267         1    99.62%
54
55 ================================================================================================
```

Hình 31: DUT: Timer Top

5.3 Exclude Check

test_bench	test_bench	Module	DU Instance	+acc=<n...	+cover=bcefst	73.57%	611	563
run_test	test_bench	Task	-	+acc=<n...	+cover=bcefst	49.28%	298	290
apb_wr	test_bench	Task	-	+acc=<n...	+cover=bcefst	83.06%	31	23
apb_rd	test_bench	Task	-	+acc=<n...	+cover=bcefst	56.32%	29	20
apb_wr_pstrb	test_bench	Task	-	+acc=<n...	+cover=bcefst	36.46%	32	23
cmp_data	test_bench	Task	-	+acc=<n...	+cover=bcefst	25.76%	11	3
reg_init_chk	test_bench	Task	-	+acc=<n...	+cover=bcefst	100.00%	16	16
cnt_chk_no_ha...	test_bench	Task	-	+acc=<n...	+cover=bcefst	0.00%	5	0
tsk_cfg_golden...	test_bench	Task	-	+acc=<n...	+cover=bcefst	100.00%	9	9
tsk_set_pslver...	test_bench	Task	-	+acc=<n...	+cover=bcefst	100.00%	3	3
tsk_set_dbg_...	test_bench	Task	-	+acc=<n...	+cover=bcefst	58.33%	3	2
tsk_rd_pslverr	test_bench	Task	-	+acc=<n...	+cover=bcefst	100.00%	1	1
u_timer	timer_top	Module	DU Instance	+acc=<n...	+cover=bcefst	100.00%	124	124
u0	apb_slave	Module	DU Instance	+acc=<n...	+cover=bcefst	100.00%	7	7
u1	reg_file	Module	DU Instance	+acc=<n...	+cover=bcefst	100.00%	103	103
u2	counter_ctrl	Module	DU Instance	+acc=<n...	+cover=bcefst	100.00%	13	13
u4	interrupt	Module	DU Instance	+acc=<n...	+cover=bcefst	100.00%	1	1
u_tb_cnt	mod_golden...	Module	DU Instance	+acc=<n...	+cover=bcefst	94.24%	20	20

Hình 32: Exclude check in Quartus

```
coverage exclude -src ../rtl/counter_ctrl.v -code b -line 33 -comment {depends on signals: div_en & timer_en & (div_val != 0), the tests are not accessed this code line directly}
coverage exclude -src ../rtl/counter_ctrl.v -code e -line 41 -comment {depends on dbg_mode which is used only for debug mode}
coverage exclude -scope /test_bench/u_timer -togglenode dbg_mode -comment {only using for debug mode}
coverage exclude -scope /test_bench/u_timer/u1 -togglenode {addr_tcr[2]} {addr_tcr[3]} {addr_tcr[4]} {addr_tcr[5]} {addr_tcr[6]} {addr_tcr[7]} {addr_tcr[12]} {addr_tcr[13]} {addr_tcr[14]} {addr_tcr[15]} -comment {Reserved bit}
coverage exclude -scope /test_bench/u_timer/u1 -togglenode {addr_tcr[16]} {addr_tcr[17]} {addr_tcr[18]} {addr_tcr[19]} {addr_tcr[20]} {addr_tcr[21]} {addr_tcr[22]} {addr_tcr[23]} {addr_tcr[24]} {addr_tcr[25]} -comment {Reserved bit}
coverage exclude -scope /test_bench/u_timer/u1 -togglenode {addr_tcr[26]} {addr_tcr[27]} {addr_tcr[28]} {addr_tcr[29]} {addr_tcr[30]} {addr_tcr[31]} {addr_thcsr[2]} {addr_thcsr[3]} {addr_thcsr[4]} {addr_thcsr[5]} {addr_thcsr[6]} {addr_thcsr[7]} {addr_thcsr[8]}
  {addr_thcsr[9]} {addr_thcsr[10]} {addr_thcsr[11]} -comment {Reserved bit}
coverage exclude -scope /test_bench/u_timer/u1 -togglenode {addr_thcsr[12]} {addr_thcsr[13]} {addr_thcsr[14]} {addr_thcsr[15]} {addr_thcsr[16]} {addr_thcsr[17]} {addr_thcsr[18]} {addr_thcsr[19]} {addr_thcsr[20]} {addr_thcsr[21]} -comment {Reserved bit}
coverage exclude -scope /test_bench/u_timer/u1 -togglenode {addr_thcsr[22]} {addr_thcsr[23]} {addr_thcsr[24]} {addr_thcsr[25]} {addr_thcsr[26]} {addr_thcsr[27]} {addr_thcsr[28]} {addr_thcsr[29]} {addr_thcsr[30]} {addr_thcsr[31]}
coverage exclude -scope /test_bench/u_timer/u1 -togglenode {addr_tier[1]} {addr_tier[2]} {addr_tier[3]} {addr_tier[4]} {addr_tier[5]} {addr_tier[6]} {addr_tier[7]} {addr_tier[8]} {addr_tier[9]} {addr_tier[10]} -comment {Reserved bit}
coverage exclude -scope /test_bench/u_timer/u1 -togglenode {addr_tier[11]} {addr_tier[12]} {addr_tier[13]} {addr_tier[14]} {addr_tier[15]} {addr_tier[16]} {addr_tier[17]} {addr_tier[18]} {addr_tier[19]} {addr_tier[20]} -comment {Reserved bit}
coverage exclude -scope /test_bench/u_timer/u1 -togglenode {addr_tier[21]} {addr_tier[22]} {addr_tier[23]} {addr_tier[24]} {addr_tier[25]} {addr_tier[26]} {addr_tier[27]} {addr_tier[28]} {addr_tier[29]} {addr_tier[30]} -comment {Reserved bit}
coverage exclude -scope /test_bench/u_timer/u1 -togglenode {addr_tier[31]} {addr_tisr[1]} {addr_tisr[2]} {addr_tisr[3]} {addr_tisr[4]} {addr_tisr[5]} {addr_tisr[6]} {addr_tisr[7]} {addr_tisr[8]} {addr_tisr[9]} {addr_tisr[10]} -comment {Reserved bit}
coverage exclude -scope /test_bench/u_timer/u1 -togglenode {addr_tisr[11]} {addr_tisr[12]} {addr_tisr[13]} {addr_tisr[14]} {addr_tisr[15]} {addr_tisr[16]} {addr_tisr[17]} {addr_tisr[18]} {addr_tisr[19]} {addr_tisr[20]} -comment {Reserved bit}
coverage exclude -scope /test_bench/u_timer/u1 -togglenode {addr_tisr[21]} {addr_tisr[22]} {addr_tisr[23]} {addr_tisr[24]} {addr_tisr[25]} {addr_tisr[26]} {addr_tisr[27]} {addr_tisr[28]} {addr_tisr[29]} {addr_tisr[30]} -comment {Reserved bit}
coverage exclude -scope /test_bench/u_timer/u1 -togglenode {addr_tisr[31]} -comment {Reserved bit}
```

Hình 33: Exclude.do file